## PART 1 - TABLE RELATIONSHIPS

For this analysis, 9 datasets were provided in Comma Separated Values format, and here is what each dataset represents:

1. Customers: contains biodata, registration information, and membership class of the business' customers.
    a. Primary Key: Customer_ID
    b. Foreign Key: *Nil*
2. Products: this is the product catalog, containing information for tracking the name, category, and value of products owned by the business.
    a. Primary Key: Product_ID
    b. Foreign Key: *Nil*
3. Orders: tracks every order made by customers, recording the number of items sold, type of items, their category, value, date the order was placed, sales channel, monetary transactions, and the completion status of each order. Every field is unique to the store the order was sent to.
    a. Primary Key: Order_ID
    b. Foreign Key: Customer_ID, Product_ID, Store_ID
4. Stores: contains information on the number of outlets, as well as their location and opening dates.
    a. Primary Key: Store_ID
    b. Foreign Key: *Nil*
5. Returns: this records all goods that were returned by customers. It shows the return date and how much was refunded to the customer.
    a. Primary Key: Return_ID
    b. Foreign Key: Order_ID
6. Employees: this is the staff list containing biodata and hiring information of all business, as well as the store outlet they work in, and their monthly renumeration.
    a. Primary Key: Employee_ID
    b. Foreign Key: Store_ID
7. Suppliers: these holds business information of all the suppliers of the business, including the goods they supply.
    a. Primary Key: Supplier_ID
    b. Foreign Key: *Nil*
8. Inventory: this keeps a record of the number of stocks available at the stores, reorder count, and the date of the last reorder.
    a. Primary Key: Inventory_ID
    b. Foreign Key: Product_ID, Store_ID
9. Order items: this holds records of the order lines.
    a. Primary Key: Order_Item_ID
    b. Foreign Key: Order_ID, Product_ID

| PARENT TABLE | CHILD TABLE | JOINED ON | RELATIONSHIP |
|---|---|---|---|
| Customers | Orders | Customer_ID | 1 : many |
| Products | Orders | Product_ID | 1 : many |
| Stores | Orders | Store_ID | 1 : many |
| Orders | Order_Items | Order_ID | 1 : 1 |
| Orders | Returns | Order_ID | 1 : many |
| Products | Inventory | Product_ID | 1 : 1 |
| Stores | Inventory | Store_ID | 1 : many |
| Stores | Employees | Store_ID | 1 : many |

*Table 1: table relationships*

## PART 8 - BUSINESS QUESTIONS

Based on the analysis conducted;

Beauty Item 210 (PROD0210) generated the highest revenue of **₦ 7,434,353.40** on the single order.

Store 003 (STR003) in Lagos has the highest performance and generated **₦ 103,227,213.70** in revenue.

Products in the Beauty category records the highest quantity of goods sold (**35,196**) and generated the highest revenue of **₦ 528,696,302.15**; making Beauty the best-selling category.

The business recorded the highest revenue in December, generating a figure of **₦ 207,785,380.80**.

Melissa Shaffer (CUST03257**)** is recorded as the customer who spends the most, as the company generated a total revenue of **₦2,057,261.55** from her orders. Other high-spending customers include;

| CUSTOMER NAME | CUSTOMER ID | REVENUE |
|---|---|---|
| Julia Moore | CUST01671 | ₦ 2,034,913.15 |
| Michael Carroll | CUST03050 | ₦ 1,981,003.15 |
| Timothy Hill | CUST02344 | ₦ 1,906,027.20 |
| Katie Peterson | CUST02253 | ₦ 1,846,925.55 |
| Mackenzie Fox | CUST01721 | ₦ 1,772,019.15 |
| Lawrence Perez | CUST04267 | ₦ 1,749,310.55 |
| Johnny Becker | CUST01248 | ₦ 1,680,352.80 |
| Rebekah Bell | CUST04068 | ₦ 1,676,470.20 |
| Melinda Ward | CUST03135 | ₦ 1,663,130.75 |

*Table 2: top 10 customers*

With a return rate of **5.23%**, products in the Home category are more frequently returned than products in other categories.

Due to an absence of a foreign key in the Suppliers table, that table cannot be connected with the other tables provided, hence, the supplier who supplies the most products cannot be determined. This limitation also made it difficult to add the product supplier name to the grand table (Orders).

21 items across 14 stores are at/below reorder level, with the most affected stores being:

| STORE ID | LOCATION | STOCK ON HAND | REORDER LEVEL | DEFICIT |
|---|---|---|---|---|
| STR013 | Lagos | 14 | 46 | 32 |
| STR009 | Ibadan | 13 | 44 | 31 |
| STR014 | Port Harcourt | 14 | 45 | 31 |
| STR022 | Awka | 12 | 41 | 29 |
| STR021 | Lagos | 21 | 47 | 26 |
| STR018 | Ibadan | 16 | 41 | 25 |
| STR006 | Kaduna | 28 | 50 | 22 |

*Table 3: stores with low inventory*

Beauty Item 479 (PROD0479) at STR017 (Abuja) needs to be urgently restocked, as it has only 10 units left against a reorder level of 29 untis.

The South region is the best performing region, recording a total revenue of **₦ 732,555,969.80**

Seeing as the Beauty category generated the highest amount of revenue and sold the highest number of products, it is advised that management invest in that category.

## PART 9 - BUSINESS RECOMMENDATIONS

**Project Title:** ShopSmart Sales Analysis
**Prepared By:** EuDoxie-Okafor Favour
**Organization:** AnalystLab Africa
**Reporting Period:** January 2025 to December 2025
**Date:** 20th October, 2026

**Executive Summary**

This report analyzes the sales operations of ShopSmart Retail Ltd., to identify business risks and opportunities for improvement.

**Findings**

1. *Order conversion is the biggest constraint on growth*. A total of 50,000 orders were placed, but only **16,529** of them were completed, **16,861** were canceled, and **16,610** were still pending at the time of analysis.
2. *Returns are majorly caused by operational factors*. 5% of orders are returned (₦61.84M refunded). Cumulatively, Late Delivery and Wrong Item cause 51% of returns, and these factors can be fixed in the business operations. Additionally, 1,685 returns were recorded against orders that were cancelled or never completed.
3. *Refund controls are weak*. The analysis revealed 270 refunds larger than the sales amount.

**Recommendations**

1. *Lost Order Recovery*. ShopSmart should implement a weekly order-status review where they follow up every pending order within 48 hours and add a mandatory cancellation-reason code at checkout and in-store. Converting just 10% of cancelled + pending order value would add about 20% revenue.
2. *Investment Areas.* The business can expand the Beauty category and promotions in the South region. They should also keep Groceries as a stable pillar, as that category has the lowest return rate (4.71%). The items below reorder level should be reordered, to keep stock flowing.
3. *Tighten returns and refunds.* Improve pick-and-pack checks to tackle sending out wrong items, and delivery SLAs to tackle late deliveries. Implementing a digital system that blocks refunds above the sale value or on cancelled orders would tackle net-negative returns.

**Risk & Opportunity**

Risk: the master data is unreliable. For example, the region labels contradict actual state regions (Lagos stores appear in more than one region) and the unit prices recorded during order transactions do not match the selling prices in the product catalog.

Opportunity: restructure the membership tiers to reflect spending. Gold, Silver and Bronze customers spend almost the same in total, so re-tiering based on actual spend and targeting the top spenders with personalized offers is a low-cost route to repeat revenue.

# APPENDIX

**Data Limitations**

1. All orders fall between 1 January and 31 December 2025, hence, no seasonal pattern can be confirmed.
2. *Region labels are unreliable*. The same state is assigned to different regions; for example, there is at least one Lagos store in the South, East, North and West. Additionally, Kano is labelled West, which is geographically wrong. Any conclusion by region should therefore be treated with caution.
3. *Sales prices do not match catalog prices*. No order's Unit_Price equals the product's Selling_Price, and 12,163 order lines are priced below Cost_Price. Additionally, the same product was sold at different prices in different outlets.
4. *Unclear field formats*. For example, it is unclear whether the values in the discount field is a percentage or a flat value.
5. Return records have contradictions.
6. Orders and Order_Items duplicate each other.
7. *Missing links between tables*. The Employees table and Orders table are not connected, making it difficult to analyze employee performance. Also, the Suppliers table cannot be linked to the Products table, making it difficult to trace products to their suppliers.
8. *Customer data quality issues*. The customer first name does not match the gender across multiple records.
9. The dataset does not contain enough operating cost to accurately determine the business profit.
10. *The Line_Total does not match the recalculated sales value*. The source Line_Total subtracts Discount as a flat value rather than treating it as a percentage.

**Assumptions**

1. Total Revenue refers to Net Revenue
2. To achieve accurate reporting of business financial position, only completed orders were used to calculate the total revenue and other financial analysis; cancelled orders are not sales, and pending orders are not yet earned.
3. The Return Rate per category is calculated using number of orders per category, and not the total number of orders.
4. The transaction price at the point of ordering is the true selling price. Revenue uses Unit_Price from Orders, not Selling_Price from Products, because Unit_Price is what the customer was actually charged. The Selling_Price from Products is assumed to be an estimated Selling_Price at the time of product purchase.
5. Each order matches only its first refund instance, as multiple returns for the same order are omitted
6. Low inventory means the available stock is at or below the reorder level.
7. Inventory is valued at cost. Inventory value = Stock_On_Hand * Cost_Price
8. Membership level is the Customer_Segment field.
9. Region is used as recorded.

## Supporting Charts
















